Electronic Circuits I (ECE 438)

Due Date: December 5th, 11.59PM

Laboratory 9

Goal

The goal of this lab is to use the MOSFETs you characterized in the previous lab to design a multi-stage amplifier to drive an LED. You will learn how to set up the correct DC bias points for an amplifier, determine current drive requirements for a LED, and characterize the small-signal AC performance of an amplifier.

Components

- Resistors as needed.
- 2x2N7000 MOSFETs (N-channel enhancement) – TO-92 package.
- 1µF/10 µF electrolytic capacitors.
- 1 Red LED and 1 Green LED as needed.

One version of the datasheet for this MOSFET may be found at Canvas

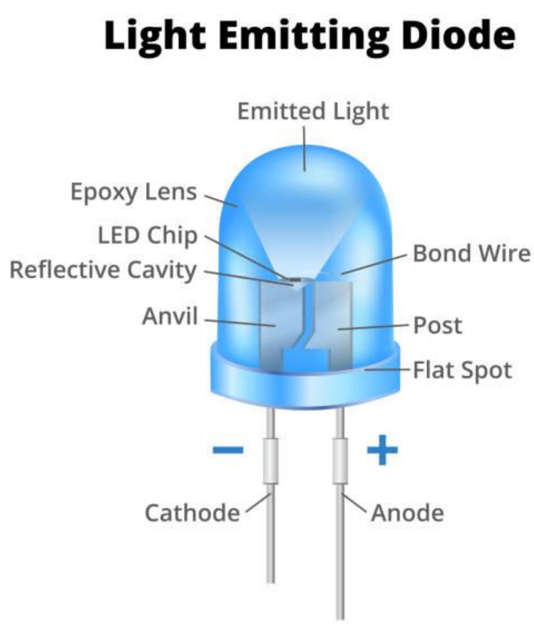

Figure 1. Schematic Diagram of Light Emitting Diode

Task 1

<u>Please keep any MOSFETS you use in this lab with you, as this a 2 week lab and label them as M1 and M2 for your lab purpose, if you use M1 at stage 2 the results might be different.</u>

The purpose of this lab is to design a two-stage linear MOS amplifier to drive an LED. In order to give you some direction in the design, you will work on the specific circuit shown in Fig 2. Each stage of the amplifier is highlighted for clarity. You will perform some reading and answer the questions that follow so that you have the necessary background material needed to complete the design. The background material is in a file on Canvas under files > Lab > Instructions > Lab 9 Related Information Fall 2025 ECE 438.

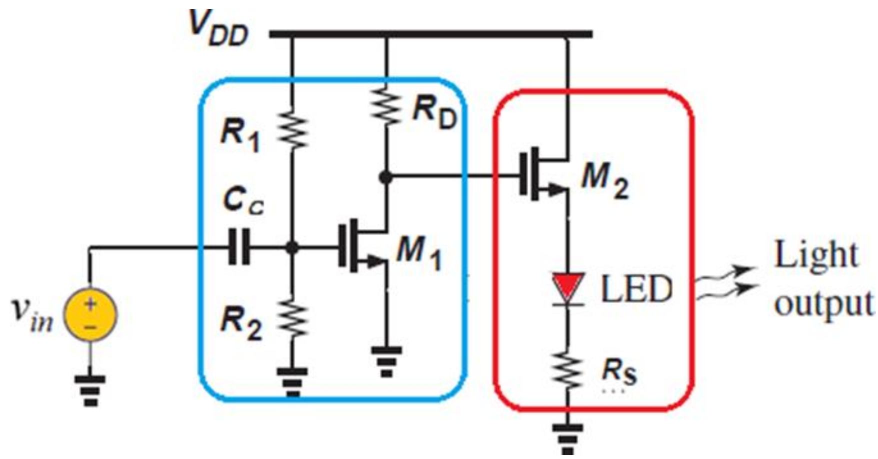

Figure 2: Two Stage MOSFET LED driver (encircled in blue is stage 1 and encircled in red is stage 2)

Read the sections corresponding to LEDs and answer the following questions.

1. What material property of a PN-junction determines the color output of a LED?
2. Identify the anode and cathode of your LEDs and set them up using a test circuit consisting of a DC voltage source and a resistor of 100 Ω.

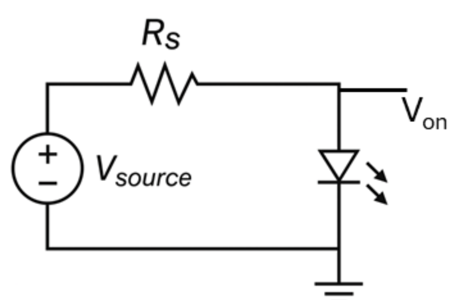

Figure 3: Setup for calculating LED voltage and Current

3. Determine a rough value of **the forward-ON** voltage of the LED for **each color** to determine the minimum anode voltage to turn it ON. Check if increasing the voltage beyond this value increases the brightness of the LED (as best as your eye can perceive), and note the voltage needed to keep the LED on with reasonable brightness. Replace the resistor with a larger value, say around **500 Ω**, or the nearest standard to see if this reduces the brightness of the LED. Why does this happen? Note down the voltage at node labelled as V_{on} and the source voltage.

Voltage for Turning on LED	Green	Red
100Ω		
500Ω		

Voltage for high brightness of LED	Green	Red
100Ω		
500Ω		

4. Calculate the DC current in the load if the output DC voltage is equal to the voltage needed to keep a red LED ON with reasonable brightness (not fully bright). **Use your answers from Task 1 step 3 above to estimate this**. Do this for resistor values 100 Ω and 500 Ω and include it in your report. (calculate this using equation: $I_{load} = V_{source} - V_{on} / R_s$)

Resistor	100Ω	500Ω
(Red) I_{load}		
(Green) I_{load}		

5. Comment on the purpose of the series resistor (whether 100 Ω or 500 Ω) in this circuit.
6. What changes do you need to make to have a comparable response from Green LED to the results of Red LED?
7. Now identify the topology of each of the two stages of the amplifier in the circuit given in Figure 2. Explain why this combination of stages may be used for driving the LED.

Task 2

Building the Circuit for Red LED

Read the section corresponding to common-drain stages and answer the following questions:

1. Perform a quick characterization as needed to obtain information about MOSFET properties (threshold voltage, operating regions etc.). You should do this by setting the circuit shown in figure 4, V_{DD} = 5V and Varying V_{in} from 0 to 5V. Identify the region where you see a swift change in V_{out}. Note down the values of Vout in that region (Vin should be 1-2V for both M1 and M2). Do this for figure 4a and 4b. Label each of the MOSFET as M2 and M1

and keep track of it, as different MOSFET can be performing in slightly different regions. Small change in the input voltage can have a large effect on the output voltage.

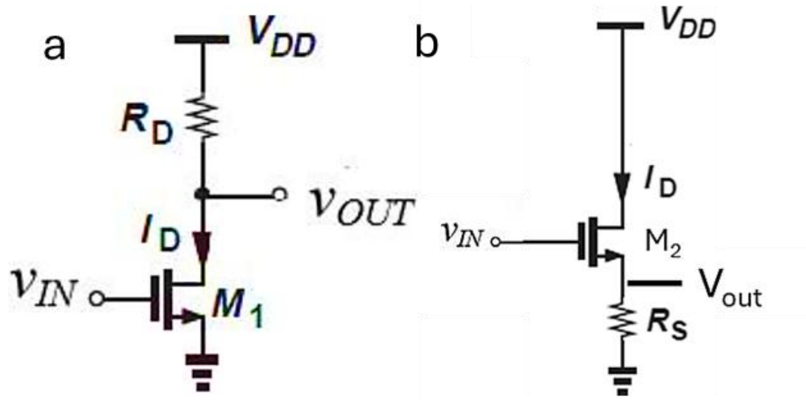

Figure 4: Circuits for characterization. Use a & b to model Stages 1 and 2 respectively.

2. Estimate what the DC bias must be on the gate of the M2 for it to output voltage required for Red LED ($V_{DD} = 5V$, $V_{out} \sim V_{source}$). Use figure 4b circuit setup to estimate this. Do this for resistor values 100 Ω and 500 Ω and include it in your report. (Suggestion: Choose V_{G2} value such that it is similar or same for both Rs, this will only effect the LED brightness than)

Resistor	100Ω	500Ω
V_{G2}		

Read the section corresponding to Common-Source gain stages, voltage divider biasing and capacitive coupling and answer the following questions:

3. If you were to design a common source amplifier with an operating point current of ~ 1 mA and an output operating point voltage equal to the voltages obtained in part (3) above ($V_{G2} = V_{out1}$), what values of drain resistor R_D would you need (for M1). Pick the nearest available standard resistor values and state them in your report.
   ($V_{DD} = 5V$) ($V_{DD} - V_{out} = I_{D1}R_D$)

Resistor	R = 100Ω	R = 500Ω
R_d		

4. Based on the current of 1mA chosen above, determine what DC gate bias is needed at the input of the Common-Source stage (M1). You should test your stage 1 circuit from Figure 4a to determine the V_{G1} value. Note that if you chose similar V_{G2} values in Step 3, you should get similar V_{G1} values also.

Voltage	R = 100Ω	R = 500Ω
V_{G1}		

5. Once you determine this voltage (V_{G1}), design the voltage divider network by calculating appropriate values of R_1 and R_2. You can freely choose these values based on the ratio you need, but respecting good design principles, choose resistor values in the range 10 kΩ to 100 kΩ.

Resistors	R = 100Ω	R = 500Ω
R_1		
R_2		

6. Calculate the small signal gain for the stage 1 and stage 2 amplifier
7. Assuming we are dealing with input sine waves in the 1KHz range, choose an appropriate value of a coupling capacitor, and justify your choice with the appropriate reasoning.

Now we will build the Two Stage LED Driver using the design parameters determined above

8. First, set up only the 1st stage of the amplifier without the input voltage, V_{in}. Measure the DC operating point of the circuit V_{GS} and V_{DS}. Tabulate and compare these values with the ones you designed for in steps 2 and 4.

Voltage	R = 100Ω	R = 500Ω
V_{GS1}		
V_{DS1}		

9. Now, set up the second part of the circuit. Measure the DC operating point of the 2nd circuit by measuring the source and drain voltage of M2.

Voltage	R = 100Ω	R = 500Ω
V_{S2}		
V_{DS2}		
V_{GS2} (Note $V_{GS2} = V_{DS1}$)		

10. If the values you measure in part (9&10) above are within 20% of what you designed, you can proceed to the next step. Otherwise, adjust your design to make sure you are within the 20% design target. State the changes you made in your report to achieve this. Make sure that $V_{S2} \sim V_{source}$ from Task 1 Step 3.

11. Apply a small sine wave (~0.2 to 0.3V_{pp})at the input of the circuit (small enough that you ensure small-signal operation) at a frequency of 1 kHz. Measure the peak-to-peak amplitude at the gate of the transistor M_1 as well as the drain and note these values in the report. Ensure for both these measurements, the signal you measure is a sine wave, and not distorted in any way. From these measurements, estimate the small-signal gain of the 1st stage. (Attach a screenshot of the Oscilloscope showing both the input and output signals)

12. To measure the gain of the complete amplifier, measure the peak-to-peak AC voltage at the gate of M1 and the peak-to-peak AC voltage at the output of the 2nd stage. From these measurements, calculate the gain in the circuit for 2nd stage. Compare the gain values to those calculated in step 7. (Attach a screenshot of the Oscilloscope showing both the input and output signals)

Task 3

Building the Circuit for Green LED

Read the section corresponding to common-drain stages and answer the following questions. These will be in many cases similar to Task 2, so it is OK to reuse your previous report structure as needed:

1. Perform a quick characterization as needed to obtain information about MOSFET properties (threshold voltage, operating regions etc.). You should do this by setting circuit shown in figure 3, VDD = 5V and Varying Vin from 0 to 5V. Identify the region where you see a swift change in Vout. You may note down the values of Vout in that region (Vin 1 to 2V). Do this for figure 4a and 4b. Label each of the MOSFET as M2 and M1 and keep track of it, as different MOSFET can be performing in slightly different regions. Small change in the input voltage can have a large effect on the output voltage.

2. Estimate what the DC bias must be on the gate of the M2 for it to output voltage required for Green LED (V_{DD} = 5V, $V_{out} \sim V_{source}$). Use figure 4b circuit setup to estimate this. Do this for resistor values 100 Ω and 500 Ω and include it in your report. (Suggestion: Choose V_{G2} value such that it is similar or close or same for both Rs, this will only effect the LED brightness than)

Resistor	100Ω	500Ω
V_{G2}		

Read the section corresponding to Common-Source gain stages, voltage divider biasing and capacitive coupling and answer the following questions:

3. If you were to design a common source amplifier with an operating point current of ~ 1 mA and an output operating point voltage equal to the voltages obtained in part (3) above ($V_{G2} = V_{out1}$), what values of drain resistor R_D would you need (for M1). Pick the nearest available standard resistor values and state them in your report.

(V_{DD} = 5V) ($V_{DD} - V_{out} = I_{D1}R_D$)

Resistor	100Ω	500Ω
R_d		

4. Based on the current of 1mA chosen above, determine what DC gate bias is needed at the input of the Common-Source stage (M1). You should test your stage 1 circuit from Figure 4a to determine the V_{G1} value. Note that if you chose similar V_{G2} values in Step 3, you should get similar V_{G1} values also

Voltage	R = 100Ω	R = 500Ω
V_{G1}		

5. Based on the current of 1mA chosen above, determine what DC gate bias is needed at the input of the Common-Source stage. Once you determine this voltage, design the voltage divider network by calculating appropriate values of R_1 and R_2. You can freely choose these values based on the ratio you need, but respecting good design principles, choose resistor values in the range 10 kΩ to 100 kΩ.

Resistors	R = 100Ω	R = 500Ω
R_1		
R_2		

6. Calculate the small signal gain for the stage 1 and stage 2 amplifiers
7. Assuming we are dealing with input sine waves in the 1KHz range, choose an appropriate value of coupling capacitor, and justify your choice with the appropriate reasoning.

Now we will build the Two Stage LED Driver using the design parameters determined above

8. First, set up only the 1st stage of the amplifier without the input voltage, V_{in}. Measure the DC operating point of the circuit V_{GS} and V_{DS}. Tabulate and compare these values with the ones you designed for in steps 3 and 4.

Voltage	R = 100Ω	R = 500Ω
V_{GS1}		
V_{DS1}		

9. Now, set up the second part of the circuit. Measure the DC operating point of the 2nd circuit by measuring the source and drain voltage of M2.

Voltage	R = 100Ω	R = 500Ω
V_{S2}		
V_{DS2}		
V_{GS2} (Note $V_{GS2} = V_{DS1}$)		

10. If the values you measure in part (9&10) above are within 20% of what you designed, you can proceed to the next step. Otherwise, adjust your design to make sure you are within the

20% design target. State the changes you made in your report to achieve this. Make sure that $V_{S2} \sim V_{source}$ from Task 1 Step 3.

11. Apply a small sine wave at the input of the circuit (small enough that you ensure small-signal operation) at a frequency of 1 kHz. Measure the peak-to-peak amplitude at the gate of the transistor M_1 as well as the drain and note these values in the report. Ensure for both these measurements, the signal you measure is a sine wave, and not distorted in any way. From these measurements, estimate the small-signal gain of the 1st stage. (Attach a screenshot of the Oscilloscope showing both the input and output signals)
12. To measure the gain of the complete amplifier, measure the peak-to-peak AC voltage at the gate of M1 and the peak-to-peak AC voltage at the output of the 2nd stage. From these measurements, calculate the gain in the circuit for 2nd stage. Compare the gain values to those calculated in step 7. (Attach a screenshot of the Oscilloscope showing both the input and output signals)

13. <u>What is the purpose of this circuit, what does it do? (Think about the input/output impedance/characteristics of the driver, and try lowering the input signal to a frequency visible to the human eye <10Hz)</u>

Report

Include a discussion at the end and conclude with your observations. Compare the results you obtained for RED and Green LED. What are the differences and similarities if any. Compare the brightness of the LED for resistor R_S between 100 Ω and 500 Ω resistor values as well as comment on the design tradeoffs, if any, encountered when deciding to choose between 100 Ω and 500 Ω (Think about the LED, output signal swing, DC operating point considerations etc.)

Some Suggestions. How should these be incorporated into instructions?

1. Aim for a reasonable LED brightness, setting VS2 to a maximum of 3.5V. Measure the LED current with a multi-meter to accurately determine M2 operating current.
2. Label the transistor/MOSFET and keep track of it. Should we ask the students to keep the MOSFETS until they are done with this lab. Or do task 1 and task 2 on first week and task 3 on second week.

I think suggestion 1 can be incorporated in Tasks 2 and 3, Steps 3, since their choosing of V_{g2} will affect the V_{s2}, and V_{s2} should be ~ V_{on} dependent on R= 100,500 ohm.

I think for suggestion 2, it is better if the MOSFETs are kept in the lab, and if they can do task 2 on first week and task 3 on second.